Design of Sequential Circuits

Flip Flop Excitation Tables:

Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

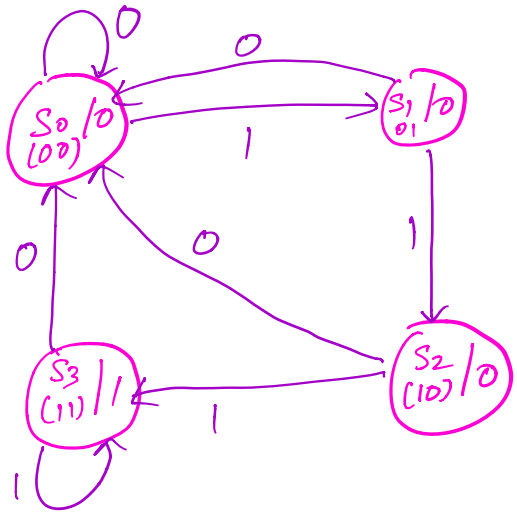
Q(t)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

Q(t)	Q(t+1)	D
0	0	0
0	1	1
1	0	0
1	1	1

Design a ckt that detects a sequence of 3 or more consecutive 1's in a string:

I/p x 0 0 1 1 1 0 1 1 1 1 0 0 0 1 1

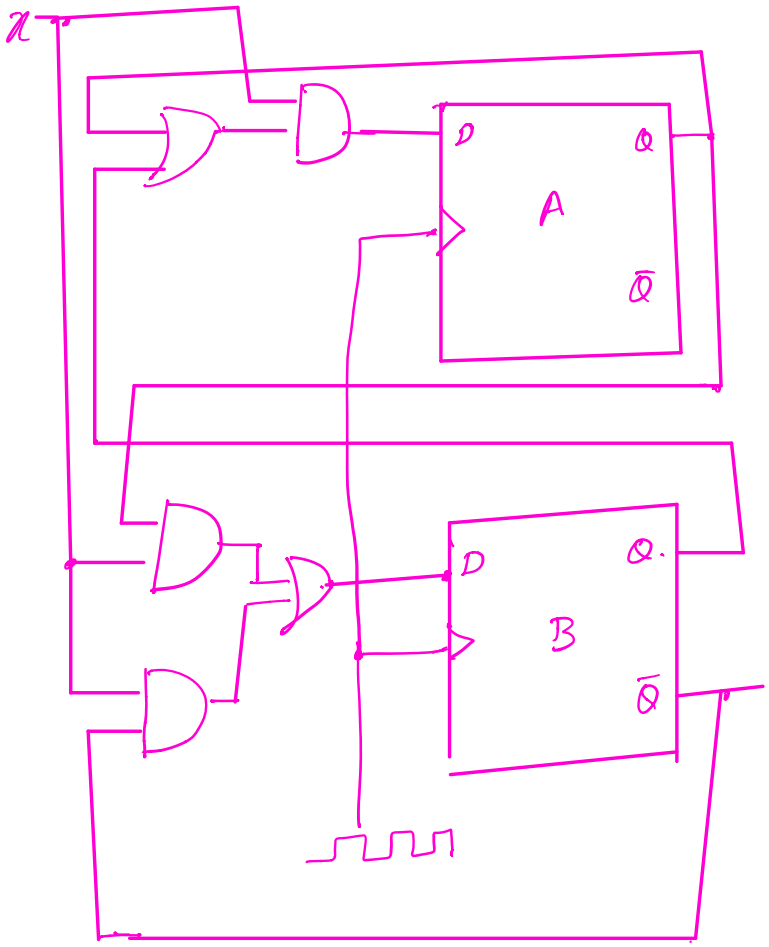
o/p y 0 0 0 0 1 0 0 0 1 1 1 0 0 0 0 0



Present State A(t)	Present State B(t)	I/P x	Next State A(t+1)	Next State B(t+1)	FF Input D_A	FF Input D_B	O/P y
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	0	0	0	0	0
0	1	1	1	0	1	0	0
1	0	0	0	0	0	0	0
1	0	1	1	1	1	1	0
1	1	0	0	0	0	0	1
1	1	1	1	1	1	1	1

$D_A(t) = \Sigma 3,5,7 = (A+B)x$

$D_B(t) = \Sigma 1,5,7 = Ax + B'x.$


x
D
Q
A
$\bar{Q}$
D
Q
B
$\bar{Q}$